

03

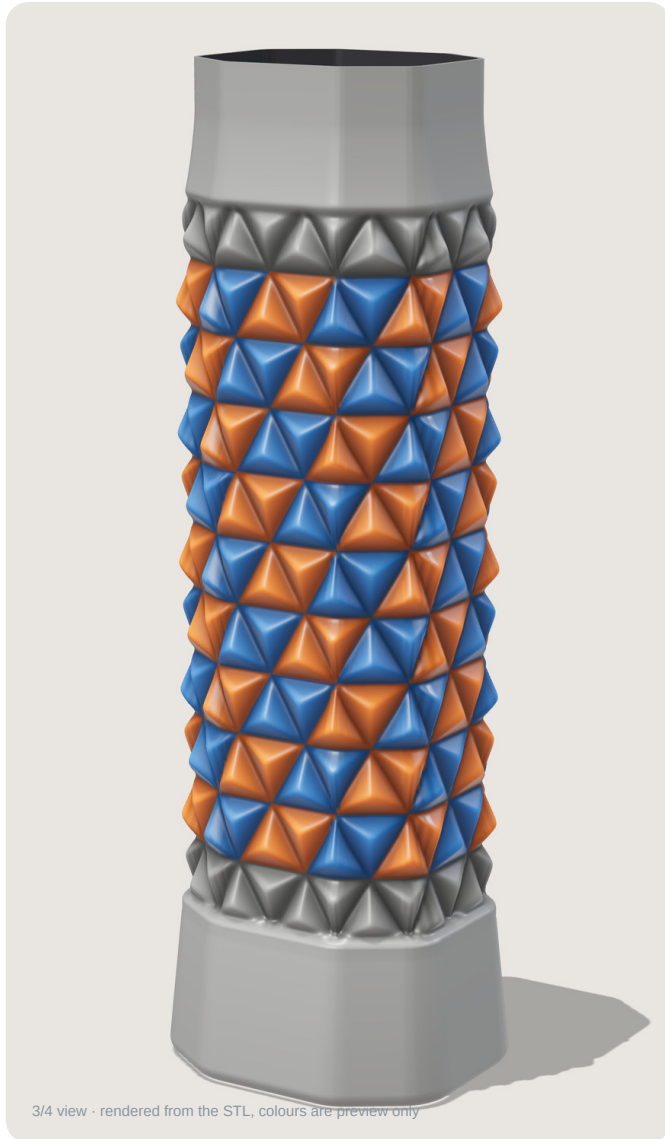
TESSEL-BLOCK

Faceted triangular pyramids in staggered rows for enhanced traction.

46.4 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	38.9 × 32.9 mm
Peak texture height	3.44 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	37.45 cm ³ / 320 cm ²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm ³		46.4 g
PETG	1.27 g/cm ³		47.6 g
TPU	1.21 g/cm ³		45.3 g
ABS	1.04 g/cm ³		38.9 g

50 g limit

COLOURWAY

#848482	#E8742A	#387AC4

3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 15.6 m (100 % flow)

DESIGN INTENT

A tessellation of triangular pyramids. Traction comes from many small angled edges rather than one deep texture, and up-pointing pyramids stand taller than down-pointing ones, giving a two-level, gem-like surface.

Facet slopes stay below 45 degrees; rim rows are flat grey bands in the colourway.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- 11 rows × 24 triangles (12 up + 12 down per row)
- Sharp V-grooves between facets
- Up-pyramids 3.44 mm, down-pyramids 3.04 mm above the bore

FILES

stl/03_tessel_block.stl	19.9 MB · 398,860 triangles
3mf/03_tessel_block.3mf	4.6 MB · colour

✓ FITMENT VERIFIED

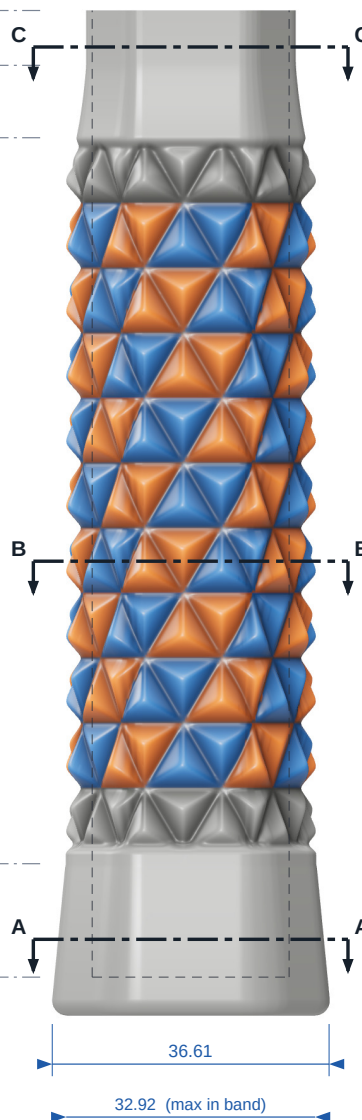
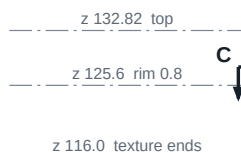
✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

✓ 3MF VALIDATED



All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

TESSEL-BLOCK

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Bore, floor and outer envelope match the reference PickleballGrip_1.3mf.
4. Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

1:1

PLA / PETG / TPU

46.4 g

REVISION

millimetres

100 %

37.45 cm³

DATE
2026-09-30

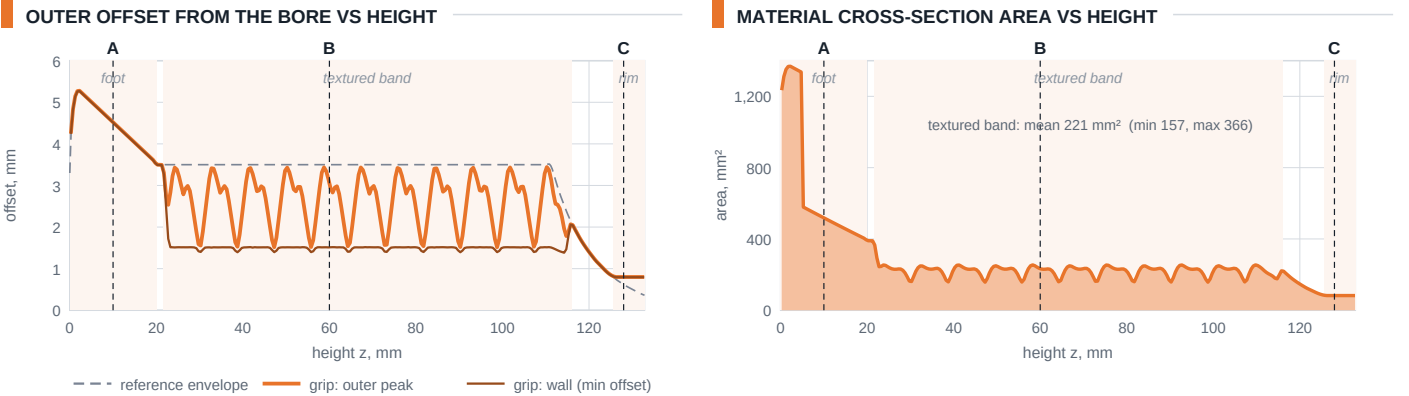
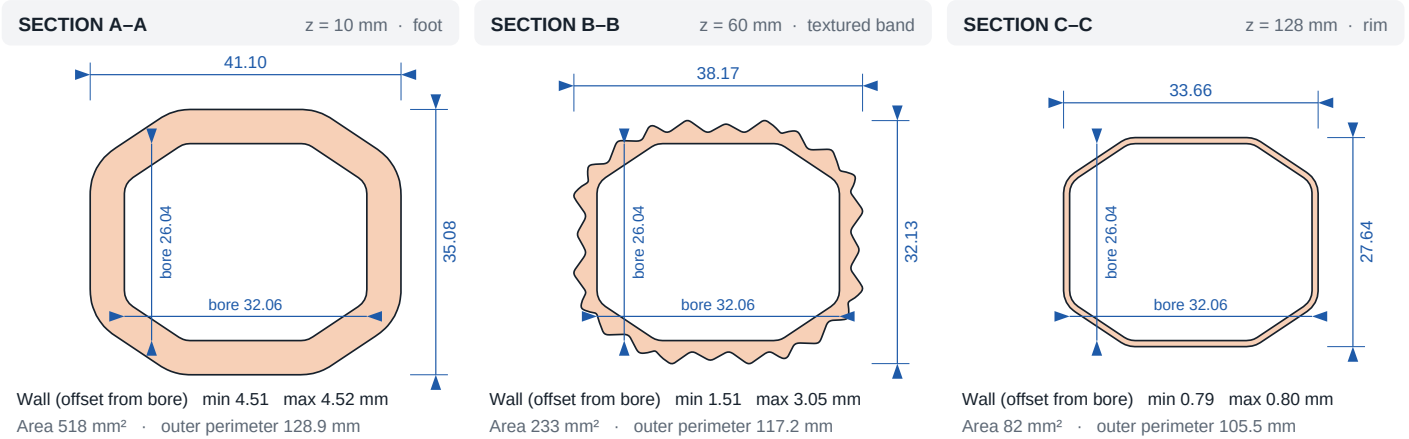
0.20 mm

320 cm²

SHEET
2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm ² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	38.94 × 32.92 mm
Circumference in band	109.9 – 120.8 mm
Volume / surface area	37.45 cm ³ / 320.4 cm ²

ZONES BY HEIGHT

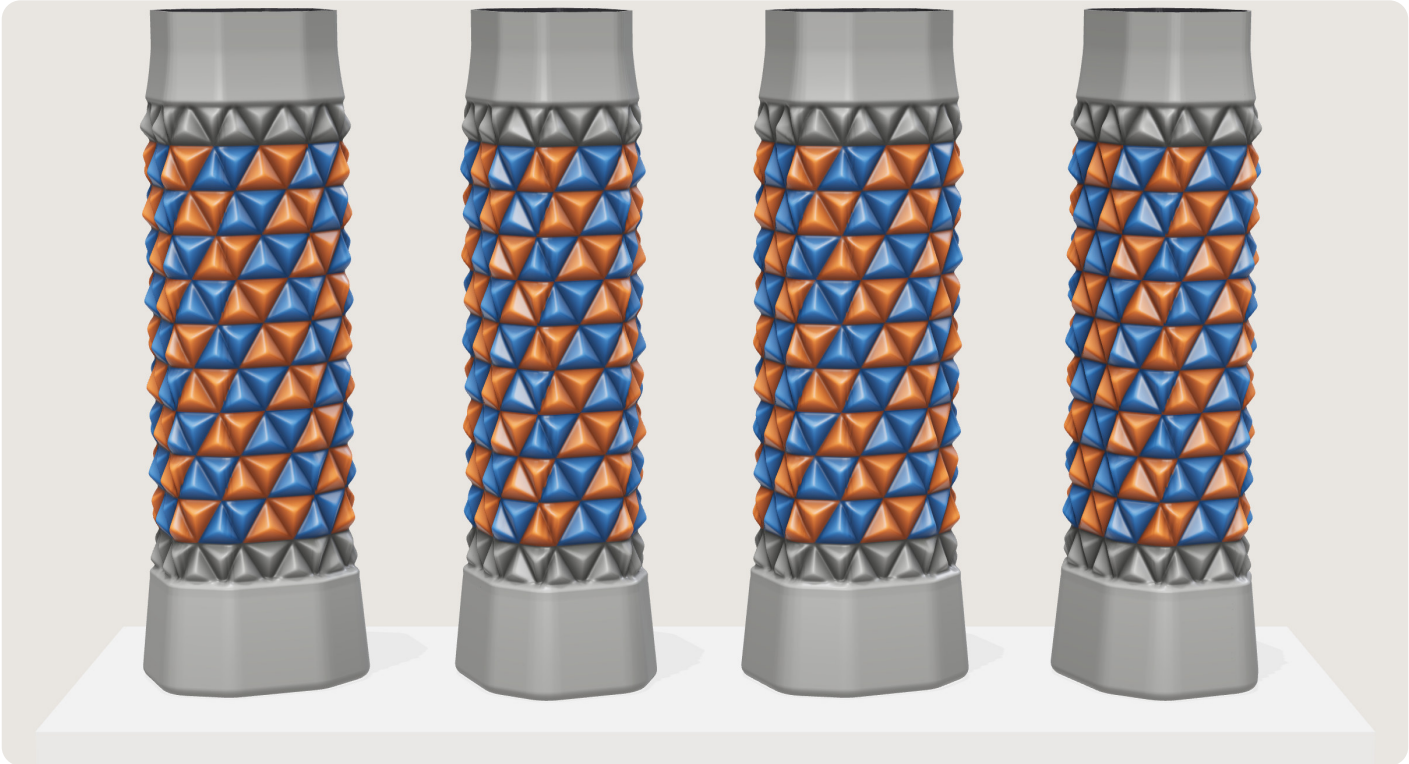
ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / 1.39 mm
Rim wall (design / measured min)			0.80 / 0.79 mm
Peak texture height above bore			3.44 mm

OUTER DIMENSIONS BY HEIGHT

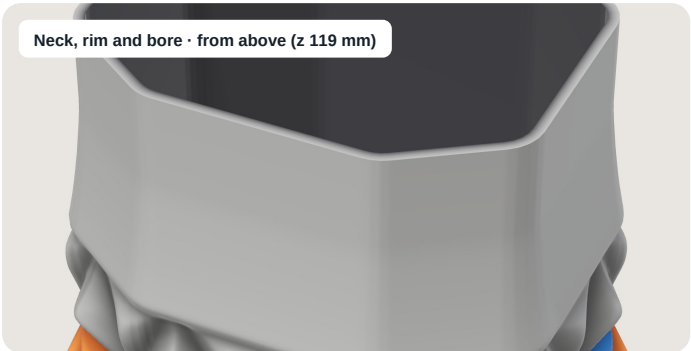
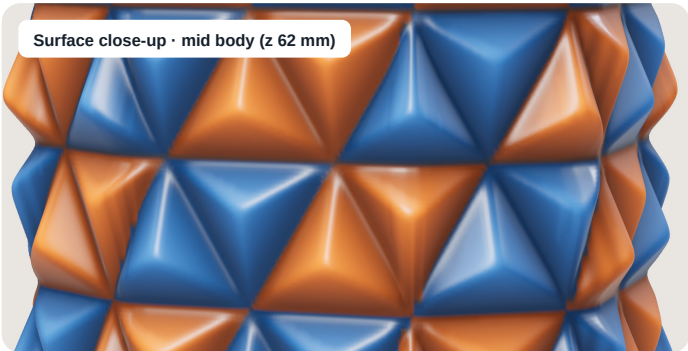
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM ²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	35.21	29.18	110.3	162	1.41	1.57
45.0	37.70	31.68	117.3	230	1.51	2.82
60.0	38.17	32.13	117.3	233	1.51	3.05
75.0	38.06	32.03	116.9	249	1.51	3.00
90.0	35.24	29.22	110.3	164	1.42	1.59
105.0	37.81	31.79	115.9	234	1.51	2.87
116.0	36.18	30.16	113.4	220	2.05	2.06
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

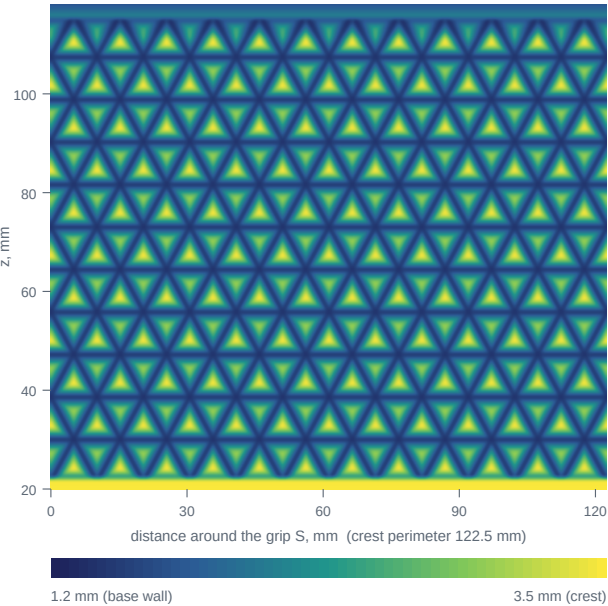
Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only



UNROLLED SURFACE · HEIGHT ABOVE BORE



DESIGN PARAMETERS

Rows	11
Triangles per row	24 (12 up + 12 down)
Triangle base	10.21 mm
Row height	8.59 mm
Up-pyramid apex	3.44 mm above bore
Down-pyramid apex	3.04 mm above bore
Groove floor	about 1.4 mm above bore

PRINT & VERIFICATION

Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none